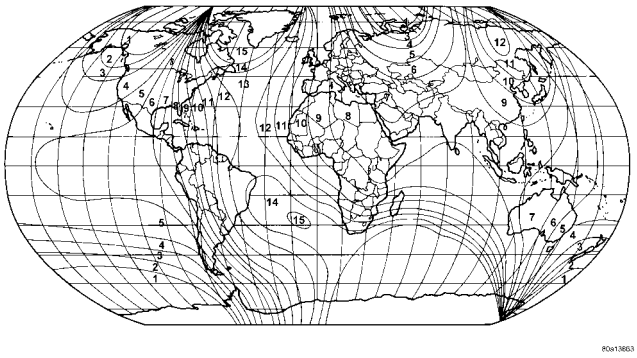

CONTINUOUS CALIBRATION

During normal operation, the EC Comp/Temp Mirror will continuously update the compass calibration to adjust for gradual changes in the vehicle's magnetic remnant field. If the vehicle is subjected to high magnetic influences, the compass may appear to indicate false headings, locked or appear unable to be calibrated. If this occurs, refer to MANUAL CALIBRATION.

COMPASS DIAGNOSTICS

If at any time the compass continually displays the incorrect direction, the reading is erratic or locked, verify the correct zone variance per COMPASS ZONE VARIANCE and manually recalibrate per MANUAL CALIBRATION. The electronic compass, although highly protected from changes in magnetic field, can be susceptible to large changes in magnetic field. Examples include, but not limited to: high tension power lines, large steel buildings, automatic car washes, large quantities of scrap metal, etc. While occurrence of this phenomenon is infrequent, it is possible.

OUTSIDE TEMPERATURE OPERATION

The EC Comp/Temp Mirror utilizes internal module memory temperature data to accurately display the outside temperature and to avoid ″hot soak″ condition readings. The displayed outside temperature information is stored within the memory of the EC Comp/Temp Mirror. The temperature memory only sets after the EC Comp/Temp Mirror has been on for 5 continuous minutes. When the EC Comp/Temp Mirror is first powered up, it retrieves the temperature data from the module memory. With the memory set when the EC Comp/Temp Mirror is powered up, the last temperature stored will be displayed and the module enters the Slow Update mode, (Slow Update = 1° increase or decrease per minute.) With the memory not set when powered up, the EC Comp/Temp Mirror will enter Fast Update mode for 5 minutes and then enter Slow Update mode, (Fast Update = Sample and display the outside air temperature every 2 seconds.)

- **TEMPERATURE UPDATE - WARM**

  On power up, when the outside temperature sensed by the ambient temperature sensor is warmer than the temperature stored in the module memory, the EC Comp/Temp Mirror will update the displayed temperature in relation to the current Update rate.

- **TEMPERATURE UPDATE - COLD**

  On power up, when the outside air temperature sensed by the ambient temperature sensor is colder than the stored memory temperature, the EC Comp/Temp Mirror will update the displayed temperature to the outside temperature at a rate of -1° every 2 seconds.

- **POTENTIAL ″ICE″ CONDITIONS**

  If the measured outside temperature is 3°C (37°F) or less, the temperature display will intermittently read the word ″ICE″ to indicate possible hazardous driving conditions. The alternating ″ICE″ display will end approximately 2 minutes after initial detection of approximately 3°C (37°F).

- **EXTREME TEMPERATURE / OPEN OR SHORT CONDITION**

  If the measured outside temperature is more than 60°C (140°F) or the ambient temperature sensor sense circuit is shorted to ground, the temp display will be ″SC″ to indicate a short circuit condition.

  If the measured outside temperature is less than -45°C (-49°F) or the ambient temperature sensor sense circuit is open, the temp display will be ″OC″ to indicate an open circuit condition.

AMBIENT TEMPERATURE SENSOR

DESCRIPTION

The ambient temperature sensor is a variable resistor that operates on a 5-volt reference signal circuit hardwired to the Compass/Temp Mirror. The outside air temperature is monitored and displayed by the Compass/Temp Mirror.

The ambient temperature sensor cannot be adjusted or repaired and, if faulty or damaged, it must be replaced.

OPERATION

The resistance in the ambient temperature sensor changes as the outside temperature rises or falls. The Compass/Temp Mirror senses the change in reference voltage through the ambient temperature sensor resistor. Based on the resistance of the ambient temperature sensor, the Compass/Temp Mirror module is programmed to correspond to a specific temperature. The Compass/Temp Mirror then displays the corresponding outside temperature received from the sensor.